

Sahen Juneja

juneja.sahen@gmail.com
linkedin.com/in/sahen-juneja/

(971)-330-6806
github.com/SJuneja0

Relocating to San Francisco CA

Education

Johns Hopkins University | Baltimore MD

M.S.E in Robotics Engineering, Track: Perception and Cognitive Systems

- Coursework: Algorithms for Sensor Based Robotics, Computer Vision, Intro to Robot Learning, Perception for Embodied AI

Worcester Polytechnic Institute | Worcester MA

BA in Robotics Engineering

Thesis: Redesign of HURON Robotic Ankle and Lower Half

- Coursework: Machine Learning, Artificial Intelligence, SLAM Robotics, Robotic Navigation, Computer Vision Robotics

Skills

Programming:	Python, C, C++, C#, Java, Git, Linux, Bash
Robotics:	ROS, VLA, Mobile Robots, Isaac Sim, Computer Vision, AI/ML
Data Analysis:	NumPy, OpenCV, PyTorch, TensorFlow
CAD:	SolidWorks, Fusion 360

Research and Publications

LCSR SPIKE Labs - Dynamic Perception: Novel Event Camera Perception System

May 2026 - Ongoing

- Researching a novel perception paradigm using dynamic cameras to track objects in robotic systems
- Developed simulation environment in Nvidia Isaac Lab to generate depth estimations to evaluate dynamic vision system
- Designed and manufactured a physical dynamic vision system with cameras that move to keep targets centered

ALMas Group - HURON: Full-size Humanoid Robot (Lower Body)

May 2023 - May 2024

- Designed a 6-DOF per leg lower half full sized bipedal robot optimizing weight, flexibility, and speed
- Used SolidWorks for simulations and assembly, Fusion for CNC code generation, and MatLab for analysis and computations and manufactured with CNC mills, waterjets, and additive manufacturing
- Ensured compatibility in electromechanical system between motors, encoders, microcontrollers, and power supply

Experience

Hologic | Marlborough, MA

R&D Mechanical Engineer Coop

Jun 2024 - Dec 2024

- Progressed concepts to prototypes for mammography device stands, internal tools, and lifecycle testing fixtures using Arduinos and Solidworks PDM
- Designed, validated, and manufactured embedded system based testing fixtures for assemblies of a mammography device
- Created and executed three testing procedures to validate products against industry quality standards

Worcester Polytechnic Institute | Worcester, MA

Fluids Lab Manager

Sep 2023 - Dec 2023

- Designed labs for undergrad students with using a Particle Image Velocimetry tank built utilizing an Arduino
- Conducted 50+ hours of testing with a self-written Python script to automate data collection of the MatLab image processing application to ensure the image capture was consistent and optimized

Worcester Polytechnic Institute | Worcester, MA

MMA President

Jan 2022 - Jan 2025

- Led two martial arts organizations, overseeing executive boards and managing activities leading to increased membership by 60 people
- Secured \$55,000 in funding for gym memberships and gear as well as hiring 4 professional martial arts instructors from various gyms to add 3 additional programs

Projects

Vision Based Robotic Manipulator Challenge

Aug 2023 - Oct 2023

- Developed a vision-based control system for an eye-in-hand robotic manipulator to achieve correct grip alignment for grasping objects
- Implemented color thresholding and feature detection algorithms to track and move several objects at a time
- Programmed Jacobians and a control loop to direct camera velocity and pose with ROS2, MatLab, Python, and OpenCV

Strategic Game AI Agent Using Search and Reinforcement Learning

Aug 2023 - Oct 2023

- Developed an AI agent in Python to play Dots and Boxes using heuristic-based strategies and alpha-beta pruning
- Implemented a minimax search algorithm to determine the optimal move within a time constraint
- Tested agent via self-play to identify weaknesses, debug, and increase speed
- Trained reinforcement learning based image recognition AI with PyTorch

Autonomous Navigation and SLAM for Mobile Robots

Jan 2022 - May 2022

- Implemented real-time A* motion planning integrated with LiDAR-based SLAM for autonomous navigation
- Designed and implemented a particle filter to address the kidnapped robot problem on a self-generated LiDAR map
- Deployed Hardware in Loop (HIL) robot in ROS with Gazebo for real-time localization and dynamic map generation

Robotic Manipulator Controls, Calibration, and Path Planning

Aug 2025 - May 2026

- Designed a modular program to draw with the UR5e using a resolved rate and IK controller to achieve sub-0.1 cm accuracy
- Utilized MATLAB Robotics and Aerospace Toolboxes with C++ and ROS2 to control a UR5e robotic manipulator, implementing forward/inverse kinematics (IK) and computing Jacobian matrices with PID controls
- Developed eye-to-hand calibration scripts with Least Squares as well as Parks and Martins algorithms using Matlab and C++
- Created an expansive search tree algorithm in C++ to path plan with the robotic manipulator, creating paths within 30 seconds that could plan through crowded workspaces

Robotic Manipulator Visualization and Testing

Jan 2023 - Feb 2023

- Deployed Hardware in Loop (HIL) tests to gauge accuracy of prototype robotic manipulators and recorded findings
- Collaborated to collect and visualize data for algorithm validation and optimization and drafted professional reports in Latex
- Calculated forward and inverse kinematics for pose control and jacobians for velocity control in MATLAB to assist in vision-based grasping task

Probabilistic Localization and Mapping for Mobile Robots

Aug 2025 - Dec 2025

- Developed localization and mapping methods for two Clearpath robots written in C++ and simulated with ROS2, Gazebo, and Rviz in Ubuntu Linux with Git
- Implemented an Extended Kalman Filter fusing GPS and IMU data to localize a Clearpath Jackal with sub-0.5 m accuracy
- Developed a particle tracking filter based on Bayesian filters using the beam model with LIDAR for SLAM on a Husky robot

Automated Machine Learning Pipeline for Image Enhancement

Aug 2025 - Dec 2025

- Designed, developed, and trained a custom deep ResUNet-based machine learning model with Pytorch, Torchvision, OpenCV, CUDA, and Numpy to enhance image exposure in both underexposed and overexposed images
- Model increased PSNR by 1.87 dB over existing research benchmarks and improved SSIM by 0.289
- Developed an automated pipeline for training, testing, and validation to support scalable multi-model development

Image Classification with Deep Neural Networks

Aug 2025 - Dec 2025

- Developed and trained a five-layer CNN using PyTorch (Torchvision) and NumPy for image classification on CIFAR-10, achieving >70% accuracy through automated hyperparameter tuning
- Implemented an automated training pipeline to evaluate combinations of epochs, dropout, learning rate, and weight decay, enabling efficient model optimization
- Implemented a CLIP-based image-text retrieval pipeline to rank and return the most semantically relevant images from CIFAR-100 using natural-language queries

Training Learning Based Robots with Pandas-Gym

Jan 2026 - May 2026

- Developed a custom Pandas-Gym simulation and to train Pick and Place, Push, and Reach tasks using the YCB dataset
- Created an expert and trained DAgger agents, reaching 100% accuracy on Push and ~70% accuracy on Pick and Place/Push
- Used Open3D and masking for point cloud detection as well as RL-zoo3 to create a custom policy for unsupervised reinforcement learning on the tasks with PPO

SLAM with Gaussian Splatting and Error-State Kalman Filters

Jan 2026 - May 2026

- Developed pipeline for Visual SLAM with Gaussian Splatting to estimate 6-DoF camera poses
- Implemented an Error-State Kalman Filter for fusing stereo images and IMU for state estimation